

AWS VPC Gateway Endpoint for S3: -

Summary: -

This task explains how to create an AWS VPC Gateway Endpoint for S3 so that communication between an EC2 instance and an S3 bucket stays inside the AWS private network instead of using the public internet. First, an EC2 instance and S3 bucket were created, and connectivity was checked using ping and traceroute commands. Then, an S3 Gateway Endpoint was created inside the VPC and attached to the route table. After enabling the endpoint, the traffic between EC2 and S3 started using the private AWS network, which improved security and performance

Let's Start: -

- Without a VPC endpoint, accessing S3 from an EC2 instance involves the public internet.
- With a VPC endpoint, the connection to S3 stays within AWS, avoiding the public internet.
- Increased security, reduced latency, and better performance
- Now For this we need one EC2 instance.

Instances (1) [Info](#)

Last updated less than a minute ago  [Connect](#) [Instance state ▼](#) [Actions ▼](#) [Launch instances ▼](#)

Saved filter sets [Choose filter set ▼](#) [< 1 >](#) 

☐	Name 	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability
☐	My-webserver	i-00a014c5906b7a133	Running  	t3.micro	3/3 checks passed	ap-south-

- I created Amazon Linux instance.
- Then I need a S3 Bucket.

General purpose buckets (203) [Info](#)

 [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

[X](#) 1 match [< 1 >](#) 

☐	Name	AWS Region	Creation date
☐	my-s3bucket2312	Asia Pacific (Mumbai) ap-south-1	May 23, 2026, 13:44:04 (UTC+05:30)

- This is my S3 Bucket.
- Open it.

Objects (1)



Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

< 1 >



☐	Name	Type	Last modified	Size	Storage class
☐	Amazon FSx.pdf	pdf	May 23, 2026, 15:29:17 (UTC+05:30)	1.4 MB	Standard

- I uploaded one document simply click on it to open.

Amazon FSx.pdf [Info](#)

Copy S3 URI

Download

Open

Object actions

[Properties](#)[Permissions](#)[Versions](#)

Object overview

Owner

ea494e9301e0e9ed79a85aa0c0282dccdbaa7eabb9b707dc9c2cfe1a489686

AWS Region

Asia Pacific (Mumbai) ap-south-1

Last modified

May 23, 2026, 15:29:17 (UTC+05:30)

Size

1.4 MB

S3 URI

s3://my-s3bucket2312/Amazon FSx.pdf

Amazon Resource Name (ARN)

arn:aws:s3::my-s3bucket2312/Amazon FSx.pdf

Entity tag (Etag)

f3eee8cbc6d2f53888f05633ebfc247b

Object URL

<https://my-s3bucket2312.s3.ap-south-1.amazonaws.com/Amazon+FSx.pdf>

- Now copy the Object URL.
- And note it in notepad.



*New Text Document (2) - Notepad

File Edit Format View Help

my-s3bucket2312.s3.ap-south-1.amazonaws.com

- Simply remove https:// from beginning and object name from last.
- Now open CMD to connect with EC2 Instance.

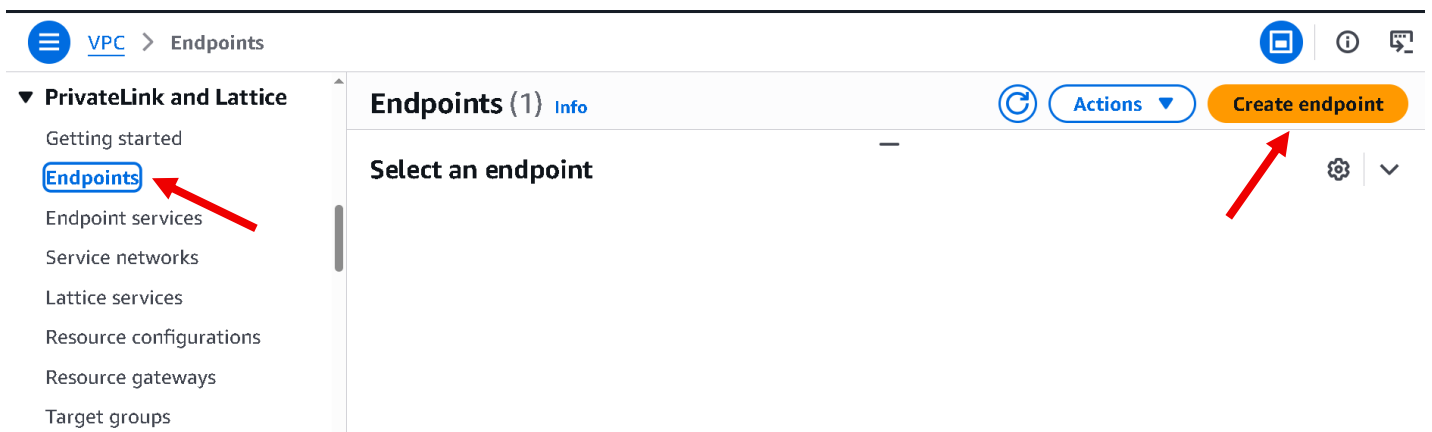
- For stop ctrl + shift + C
- right now, communication is happening.
- Now if communication is happening, it can go through the internet or maybe Amazon private network.
- Now we don't know this right now.

```

[ec2-user@ip-172-31-45-176 ~]$ traceroute my-s3bucket2312.s3.ap-south-1.amazonaws.com
traceroute to my-s3bucket2312.s3.ap-south-1.amazonaws.com (52.219.158.78), 30 hops max, 60
byte packets
 1  244.5.0.51 (244.5.0.51)  1.040 ms 244.5.0.43 (244.5.0.43)  7.014 ms *
 2  100.65.19.160 (100.65.19.160)  1.552 ms 100.65.18.144 (100.65.18.144)  5.140 ms 100.65
.20.240 (100.65.20.240)  6.918 ms
 3  100.66.9.98 (100.66.9.98)  6.819 ms 100.66.8.112 (100.66.8.112)  2.917 ms *
 4  100.66.11.162 (100.66.11.162)  8.656 ms 100.66.10.66 (100.66.10.66)  6.847 ms 100.66.1
1.2 (100.66.11.2)  1.006 ms
 5  100.66.15.71 (100.66.15.71)  3.122 ms 100.66.15.33 (100.66.15.33)  4.913 ms *
 6  100.66.12.67 (100.66.12.67)  3.690 ms * *
 7  100.65.24.241 (100.65.24.241)  6.704 ms 100.65.24.113 (100.65.24.113)  6.670 ms 100.65
.25.113 (100.65.25.113)  6.669 ms
 8  s3-r-w.ap-south-1.amazonaws.com (52.219.158.78)  0.225 ms 0.240 ms 0.223 ms
[ec2-user@ip-172-31-45-176 ~]$

```

- Enter traceroute Bucket URL
- if you look at this IP address, it is going through the internet.
- Why? It is going through the internet?
- Because right now default communication is actually happening.
- I have no end point.
- So, now I am going to VPC and open Endpoint.
- And create Endpoint.



- Now click Create Endpoint.

Endpoint settings

Specify a name and select the type of endpoint.

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

S3Endpoint

Type

Select a category

☒ AWS services

Connect to services provided by Amazon with an Interface endpoint, or a Gateway endpoint

☐ PrivateLink Ready partner services

Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink

☐ AWS Marketplace services

Connect to SaaS services that you have purchased through AWS Marketplace with an Interface Endpoint

☐ EC2 Instance Connect Endpoint

An elastic network interface that allows you to connect to resources in a private subnet

☐ Resources

Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink

☐ Service networks

Connect to VPC Lattice service networks with a Service network endpoint. Uses AWS PrivateLink

- Enter name.
- Type AWS Service.
- Scroll down

Services (1/8)

Search 

S3  Clear filters

< 1 > 

	Service Name	Owner	Type
☐	aws.api.ap-south-1.s3files	amazon	Interface
☒	com.amazonaws.ap-south-1.s3	amazon	Gateway 
☐	com.amazonaws.ap-south-1.s3	amazon	Interface
☐	com.amazonaws.ap-south-1.s3-outposts	amazon	Interface
☐	com.amazonaws.ap-south-1.s3express	amazon	Gateway
☐	com.amazonaws.ap-south-1.s3tables	amazon	Interface
☐	com.amazonaws.ap-south-1.s3vectors	amazon	Interface
☐	com.amazonaws.s3-global.accesspoint	amazon	Interface

- Search S3 and Select Gateway Type.
- Scroll down.

Network settings

Select the VPC in which to create the endpoint

VPC

Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-072c4acd761c3b942  

► Additional settings

- Now select default VPC.
- Scroll down.

Route tables (1/1) [Info](#)

Search

< 1 > 

☒	Name	Route Table ID	Main	Associated Id
☒ 		rtb-017299725a38d5666	Yes	2 subnets

 When you use an endpoint, the source IP addresses from your instances in your affected subnets for accessing the AWS service in the same region will be private IP addresses, not public IP addresses. Existing connections from your affected subnets to the AWS service that use public IP addresses may be dropped. Ensure that you don't have critical tasks running when you create or modify an endpoint.

rtb-017299725a38d5666 

- Now select the route table.
- Scroll down.

Policy [Info](#)

VPC endpoint policy controls access to the service.

- ☒ **Full access** 
- Allow access by any user or service within the VPC using credentials from any Amazon Web Services accounts to any resources in this Amazon Web Services service. All policies — IAM user policies, VPC endpoint policies, and Amazon Web Services service-specific policies (e.g. Amazon S3 bucket policies, any S3 ACL policies) — must grant the necessary permissions for access to succeed.
- ☐ **Custom**
- Use the [policy creation tool](#) to generate a policy, then paste the generated policy below.

1	
---	--

- Select full access.

- scroll down

Tags

Key

Value - optional

Remove

Add new tag

You can add 49 more tags.

Cancel

Create endpoint

- Simply click Create endpoint.

vpce-0295eb8e78e313c38 / S3Endpoint

Actions ▾

Details

Endpoint ID

 vpce-0295eb8e78e313c38

VPC ID

 vpc-072c4acd761c3b942

DNS record IP type

service-defined

Private DNS specified domains

—

Status

 Available

Status message

—

IP address type

ipv4

Creation time

Saturday, 23 May 2026 at 16:09:21 GMT+5:30

Service name

 com.amazonaws.ap-south-1.s3

Service region

ap-south-1

Endpoint type

Gateway

Private DNS names enabled

No

Private DNS preference

—

- It's Available.
- Go to CMD and check again Enter ping Bucket URL

```

[ec2-user@ip-172-31-45-176 ~]$ ping my-s3bucket2312.s3.ap-south-1.amazonaws.com
PING s3-r-w.ap-south-1.amazonaws.com (16.12.40.82) 56(84) bytes of data.
64 bytes from s3-r-w.ap-south-1.amazonaws.com (16.12.40.82): icmp_seq=1 ttl=248 time=0.457 ms
64 bytes from s3-r-w.ap-south-1.amazonaws.com (16.12.40.82): icmp_seq=2 ttl=248 time=0.453 ms
64 bytes from s3-r-w.ap-south-1.amazonaws.com (16.12.40.82): icmp_seq=3 ttl=248 time=0.458 ms
^C
--- s3-r-w.ap-south-1.amazonaws.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.453/0.456/0.458/0.002 ms
[ec2-user@ip-172-31-45-176 ~]$ traceroute my-s3bucket2312.s3.ap-south-1.amazonaws.com
traceroute to my-s3bucket2312.s3.ap-south-1.amazonaws.com (52.219.156.74), 30 hops max, 60
byte packets
 1  * * *
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  *
```

- Now we trace this we see changes.
- It is actually going through the private Amazon network.
- And you cannot trace this.